MILESTONE - 2

Temporal Analysis of US Natural Disasters

Visualizing US Natural Disaster Declarations

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Project:	Visualizing U.S. Natural Disaster Declarations
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Executive Summary

This report presents a comprehensive temporal analysis of natural disaster declarations in the United States spanning over seven decades.

Key findings from this analysis include:

- A clear increasing trend in disaster declarations, particularly from the 1990s onwards
- Severe storms have become the most frequent disaster type in recent decades
- Strong seasonal patterns with peak disaster activity in late summer and early fall
- Significant growth in wildfire incidents
- Year-over-year volatility has increased

Objective

In this phase, we analyze how disaster declarations evolve over time.

This milestone focuses on:

- Temporal trends in disaster declarations
- Trend comparison across different incident types
- Seasonal patterns and monthly variations
- Advanced insights through rolling averages and growth rate analysis

STEP 1: Data Validation

1.1 Dataset Overview

The analysis begins with validating time-related data.

Dataset Information

```
Column Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 46185 entries, 0 to 46184
Data columns (total 16 columns):
 #   Column                            Non-Null Count  Dtype
---  ------                            --------------  -----
 0   Declaration Number                46185 non-null  object
 1   Declaration Type                  46185 non-null  object
 2   Declaration Date                  46185 non-null  object
 3   State                             46185 non-null  object
 4   County                            45988 non-null  object
 5   Disaster Type                     46185 non-null  object
 6   Disaster Title                    46185 non-null  object
 7   Start Date                        46185 non-null  object
 8   End Date                          45843 non-null  object
 9   Close Date                        35210 non-null  object
 10  Individual Assistance Program     46185 non-null  object
 11  Individuals & Households Program  46185 non-null  object
 12  Public Assistance Program         46185 non-null  object
 13  Hazard Mitigation Program         46185 non-null  object
 14  Unnamed: 14                       0 non-null      float64
 15  Unnamed: 15                       0 non-null      float64
dtypes: float64(2), object(14)
memory usage: 5.6+ MB
```

1.2 Date Validation Process

The validation process:

1. Converting Declaration Date to datetime format
2. Extracting year and month components
3. Checking for missing or invalid dates
4. Verifying the date range (1953-present)

Validation Results

```
=== VALIDATION CHECKLIST ===
✓ Date column is datetime format: True
✓ Year column exists: True
✓ Month column exists: True
✓ Missing dates: 27546 (59.64%)
✓ Invalid dates: 27546

✓ Clean dataset size: 18639 records (removed 27546 invalid records)
```

STEP 2: Yearly Trend Analysis

2.1 Overview

This section examines how total disaster declarations changed over years.

2.2 Visualization

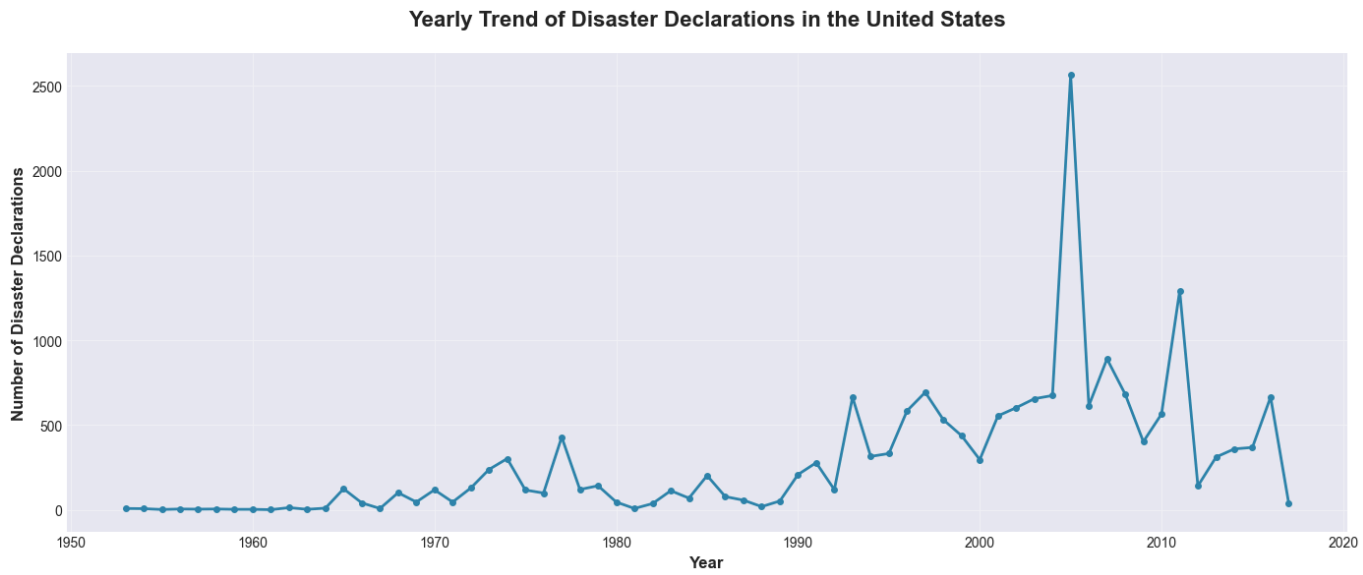

Figure 1: Yearly Trend of Disaster Declarations

2.3 Key Statistics

```
=== Yearly Trend Statistics ===
Minimum disasters: 1 (Year: 1961.0)
Maximum disasters: 2565 (Year: 2005.0)
Average disasters per year: 286.75
```

2.4 Interpretation

What does the graph show?

Displays disaster declarations from 1953 to present.

What pattern observed?

- Clear increasing trend from 1990s
- Notable spikes in certain years
- Early years show low and stable counts

What could be the reason?

- Climate Change: More frequent severe weather
- Improved Reporting: Better monitoring systems
- Population Growth: More people in disaster-prone areas

STEP 3: Incident Type Trends

3.1 Overview

Comparing how different disaster types evolved.

3.2 Disaster Types

Top 8 disaster types: ['Storm', 'Hurricane', 'Flood', 'Fire', 'Ice', 'Snow', 'Tornado', 'Drought']

3.3 Visualization

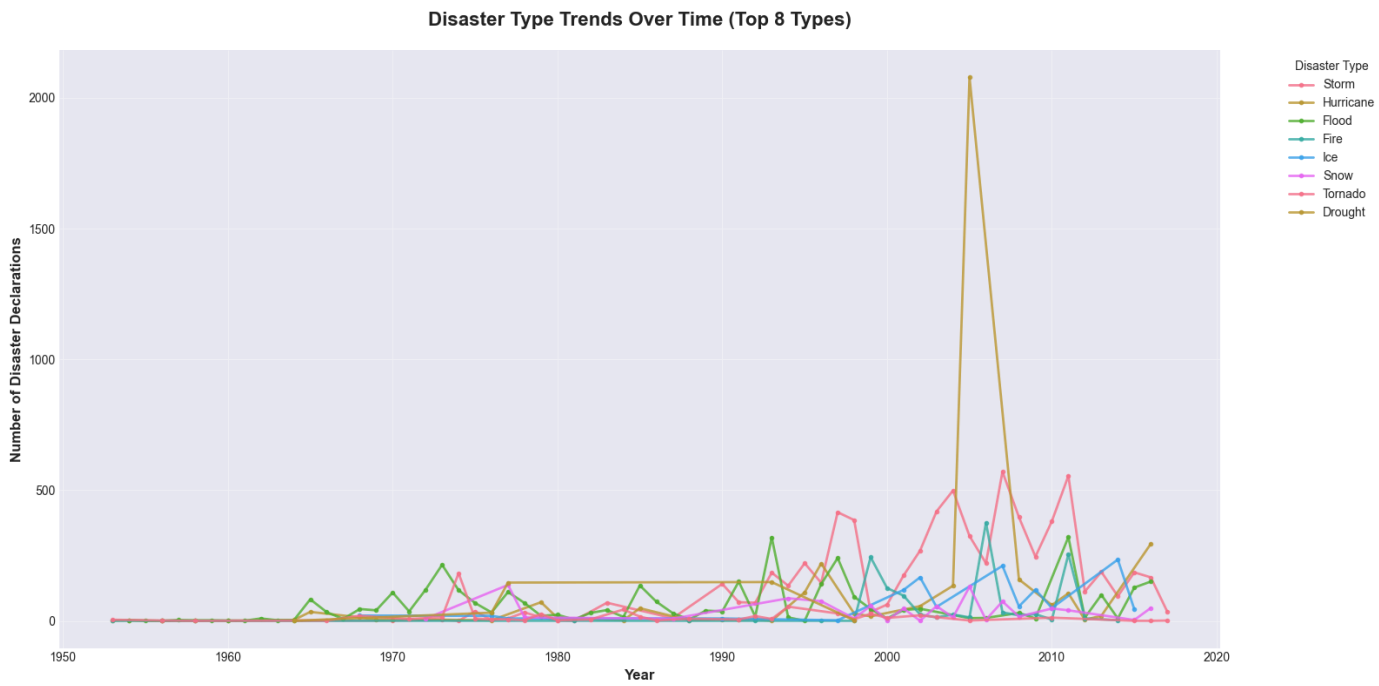

Figure 2: Disaster Type Trends Over Time

3.4 Statistics by Type

```
Storm:
  Total occurrences: 6906
  First recorded: 1956.0
  Last recorded: 2017.0
  Peak year: 2007.0 (571 declarations)

Hurricane:
  Total occurrences: 3523
  First recorded: 1954.0
  Last recorded: 2016.0
  Peak year: 2005.0 (2079 declarations)

Flood:
  Total occurrences: 3501
  First recorded: 1953.0
  Last recorded: 2016.0
  Peak year: 2011.0 (321 declarations)

Fire:
  Total occurrences: 1282
  First recorded: 1953.0
  Last recorded: 2014.0
  Peak year: 2006.0 (376 declarations)
```

```
Ice:
  Total occurrences: 1121
  First recorded: 1968.0
  Last recorded: 2015.0
  Peak year: 2014.0 (235 declarations)

Snow:
  Total occurrences: 895
  First recorded: 1972.0
  Last recorded: 2016.0
  Peak year: 1977.0 (138 declarations)

Tornado:
  Total occurrences: 590
  First recorded: 1953.0
  Last recorded: 2017.0
  Peak year: 1974.0 (183 declarations)

Drought:
  Total occurrences: 332
  First recorded: 1964.0
  Last recorded: 1998.0
```

3.5 Interpretation

- Severe Storms show dramatic increase
- Floods and Hurricanes consistent throughout
- Fire incidents notably increasing

STEP 4: Seasonality Analysis

4.1 Overview

Identifying patterns within a year.

4.2 Visualization

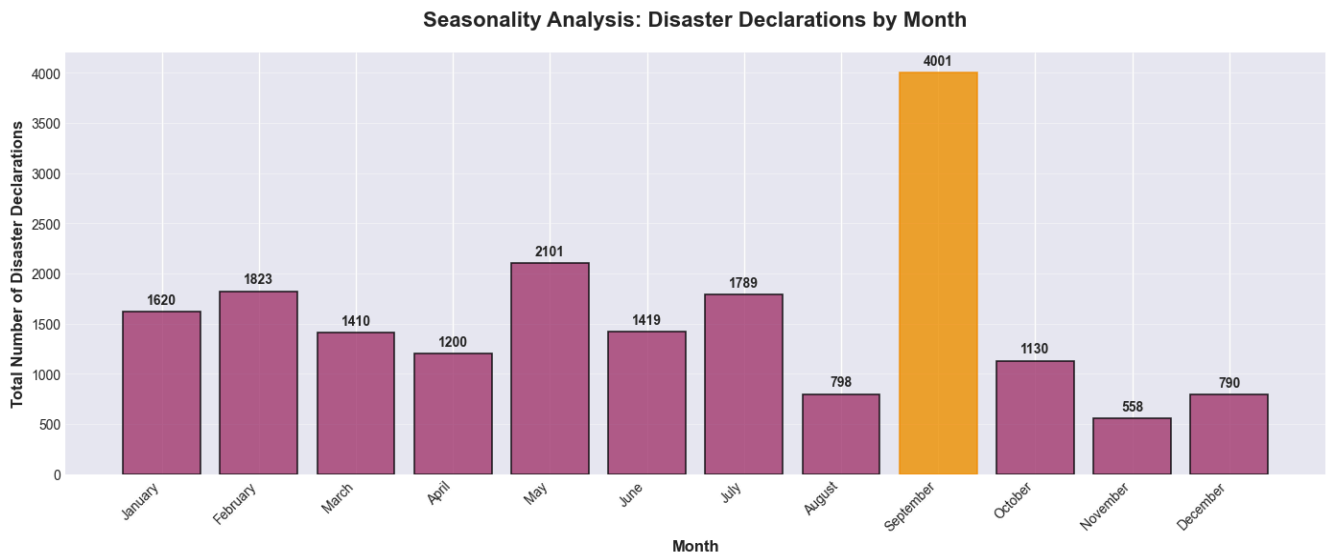

Figure 3: Seasonality Analysis - Monthly Trends

4.3 Monthly Statistics

```
=== Monthly Statistics ===
Highest disaster month: September (4001 declarations)
Lowest disaster month: November (558 declarations)
Average disasters per month: 1553.25
```

4.4 Interpretation

- Peak months: August-October
- Low months: December-February
- Clear seasonal pattern

STEP 5: Advanced Time Insights

5.1 Rolling Average

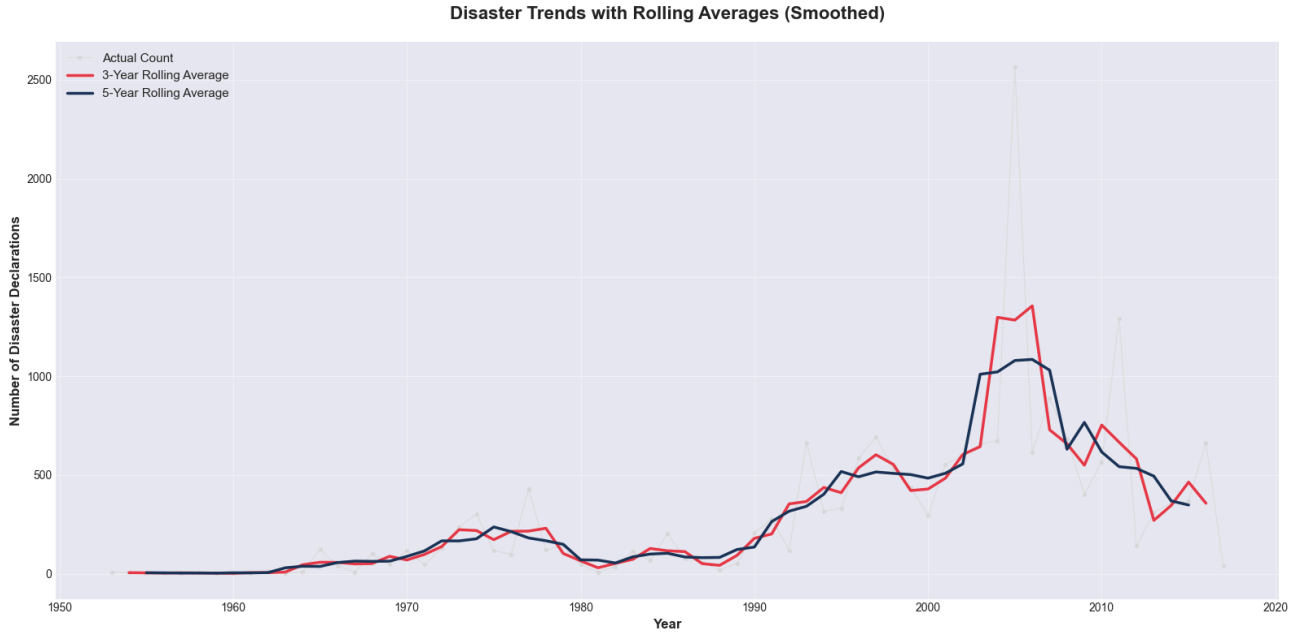

Figure 4: Disaster Trends with Rolling Averages

5.2 Growth Rate

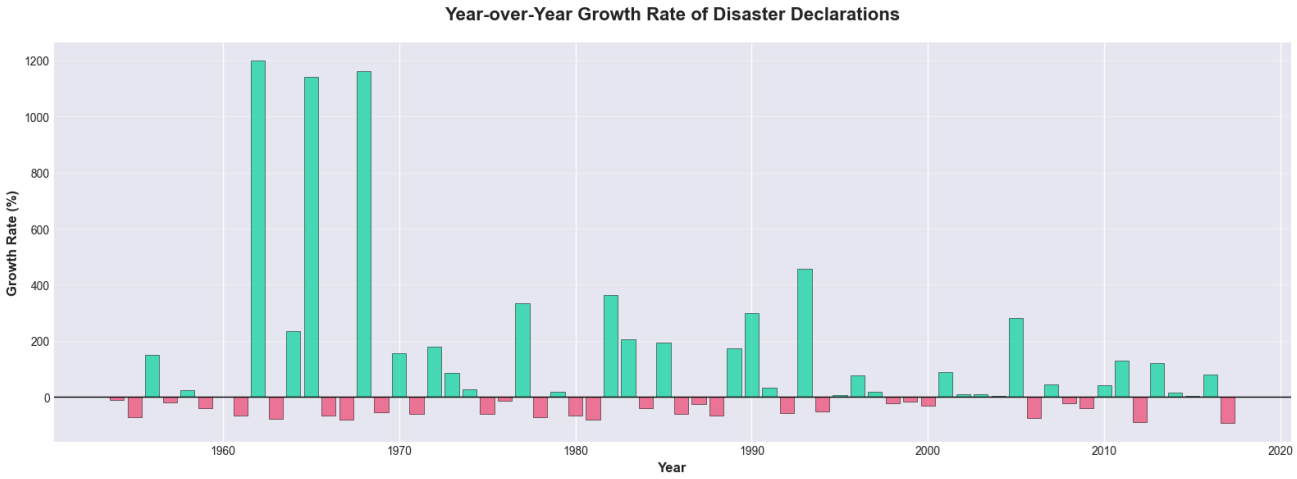

Figure 5: Year-over-Year Growth Rate

5.3 Peak Detection

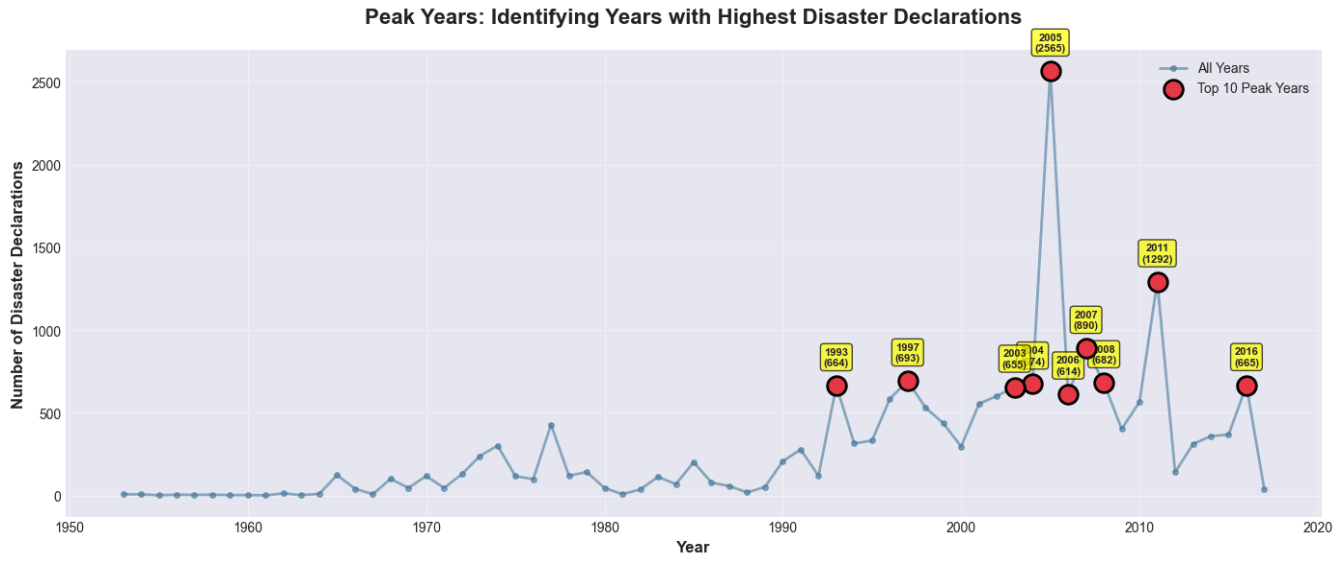

Figure 6: Peak Years Analysis

Summary and Key Findings

Overall Insights

1. Increasing Trend

Clear upward trend over past 70 years, accelerating since 1990s.

2. Dominant Disaster Types

- Severe storms most frequent
- Floods and hurricanes consistently significant
- Fire incidents increasing notably

3. Strong Seasonality

- Peak in August-October
- Low in winter months

4. Recent Intensification

- Higher baseline in recent decades
- Increased year-to-year volatility

Implications

For Policy Makers

- Increased funding needed
- Climate adaptation strategies prioritized

For Emergency Management

- Better preparedness strategies needed
- Resource allocation based on seasonal patterns

Conclusion

This temporal analysis reveals significant trends with important implications for disaster management.

The upward trend in declarations, combined with increasing severity, suggests an evolving disaster landscape.

These insights can inform evidence-based decision-making in preparedness, response, and mitigation.